

What Do Language Models Know About the Past?

Probing base model probabilities for occupational structure and prestige, United States, 1850--1940

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1 The problem

The collective mental worlds of the past aren't often preserved in archives. How much prestige was attached to a given occupation is a type of cultural knowledge that is shared by contemporaries but often not explicitly recorded. Unable to survey the dead, this knowledge is lost to historians. Language models are trained on the textual residue of those worlds. If collective and perspectival knowledge is encoded anywhere, it may be encoded there, distributed across documents.

Can language models recover collective cultural knowledge from the past?

2 Design

Two-stage design. The first recovers recorded facts, so the method can be checked against them. The second applies the same method where no record exists.

Task 1 — documented. Occupation distributions, 1850–2024. Ground truth: IPUMS census. *Does the probe recover what we can check?*

Task 2 — undocumented. Prestige hierarchies, 1850–1940. No survey exists before 1925. *Can models produce historically plausible prestige classifications?*

3 Method

The prompt

In {year}, the {role} is {name}. A popular {media_type} is {media}. {event}.

Structure: A common occupation in the United States was...

Prestige: An occupation near the top of the status hierarchy in the United States is...

1910: Postmaster General Frank Harris Hitchcock • *The Kingdom of Slender Swords* • garment workers' strikes

- Anchor.** Period-appropriate officeholder, popular work, and event fill the slots. No year-specific outcome is named.
- Score.** Log-probability of every completion in a fixed occupation slate.
- Pool and normalise.** Variants summed within group, normalised to a distribution over the slate.
- Compare.** Spearman rank correlation against ground truth.

Seven open-weight base models. OLMo-2-32B • Qwen3-14B • Qwen3.5-0.8B • Mistral-7B • gemma-3-27b-pt • falcon-40b • **talkie-1930-13b** as a bounded model comparison.

Ground truth. *Occupational structure:* IPUMS census. • *Prestige:* Seven historical classifications, coded into six ordinal tiers; HISCAM for the UK.

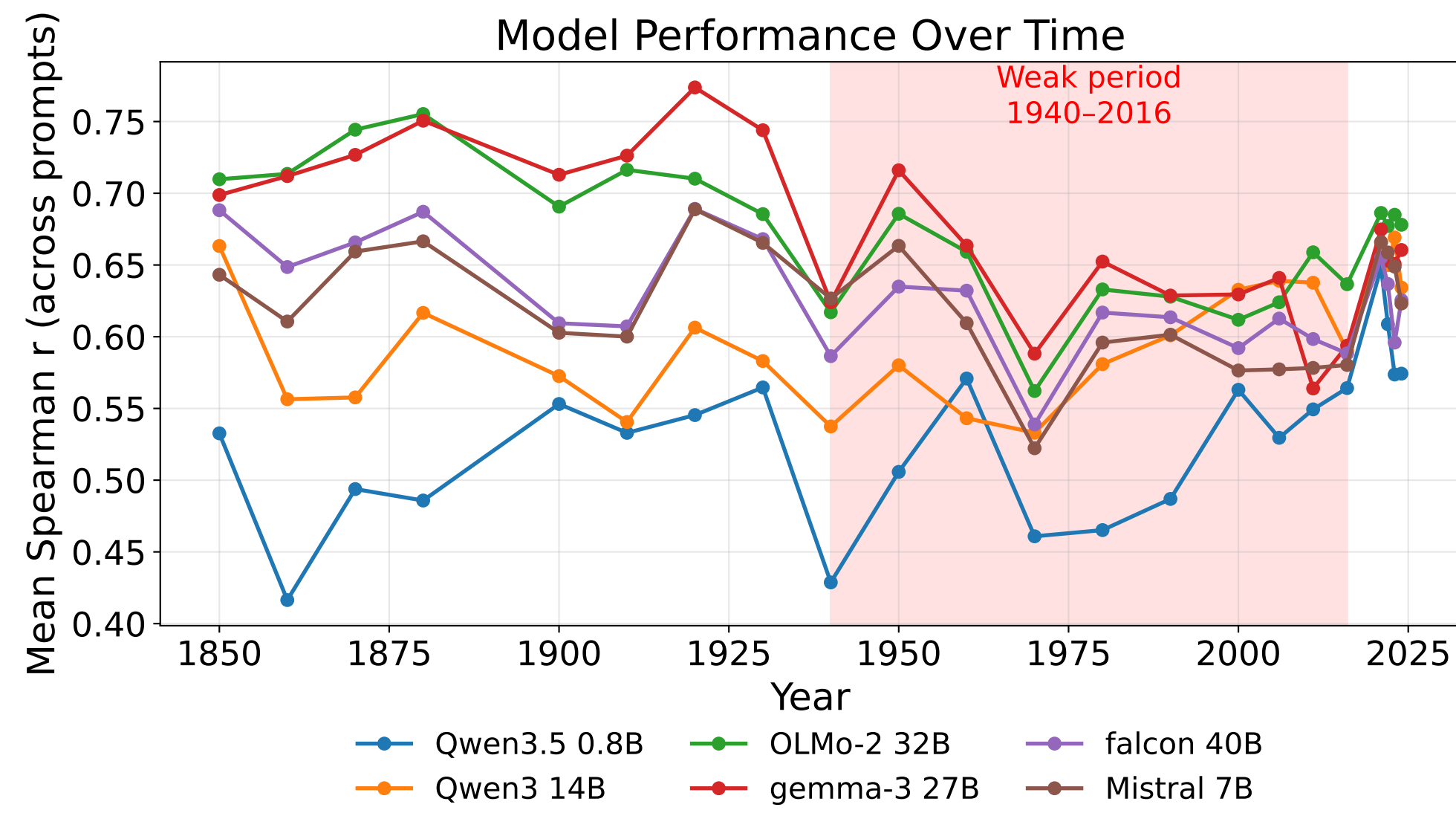
7 Takeaways

- Probe probabilities, do not parse text.** Generation fails at this task in every base model we tested. Models carry a recoverable ranking in their probability distribution.
- Base models, not aligned ones.** Alignment training distorts the historical signal, whereas base models produce better recovery.
- Recovered mentalities are perspectival.** What is recovered depends on the vantage point prompts encode — which is as much of a substantive finding than a methodological one.

What this is for. A working pipeline and a set of heuristics for asking whether collective historical fact can be recovered from language models at all. Occupational prestige is the test case. However, this method generalises to other widely held, rarely recorded collective mental content.

4 Task 1: Occupational Structure

With no temporal cue, models produce a generic distribution with a moderate mean correlation with census occupational distributions ($r = 0.485$). Adding a year marker lifts correlation to 0.630. Adding cultural anchors lowers mean correlations slightly to 0.620, but improves recovery for 19th and early 20th century periods.



Anchoring changes where recovery happens Mean Spearman r against census structure, averaged across prompts. Cultural anchoring gains ≈ 0.03 per decade from 1880–1940 and loses as much from 1940–2016 on. The weak period is where period framing has least to add, as late twentieth-century labour force is already the model's default.

Not all conditioning helps.

Condition	Matched r	vs. null
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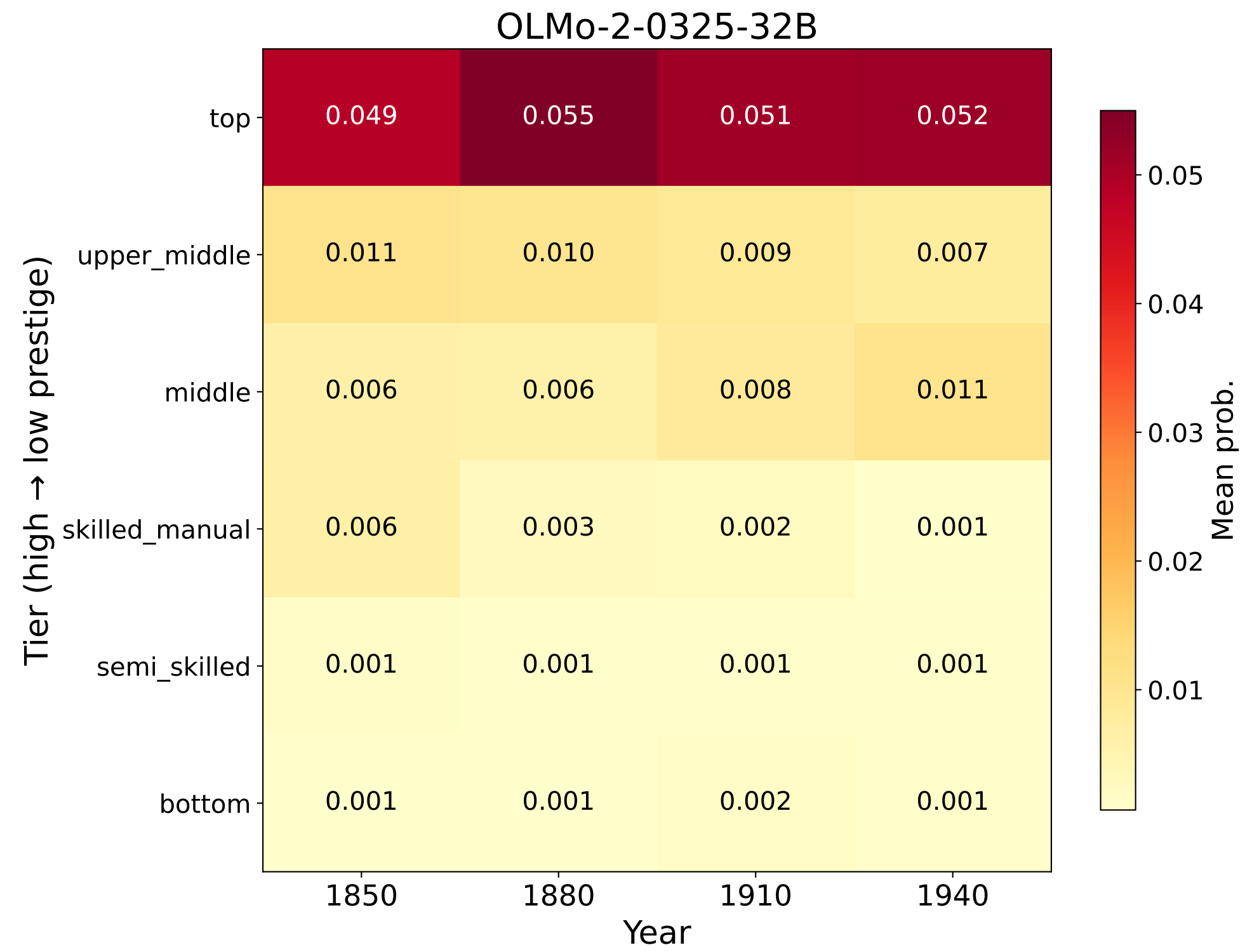
No year (null)	0.485	—
Year only	0.630	+0.145
Year + cultural anchors	0.620	+0.135
Year + period slang	0.572	+0.087
Year repeated	0.576	+0.091

Slang degrades recovery for all six models ($d = -0.90$ to -1.97 , $p < .001$); repetition for five of six. Anchoring is the only manipulation that holds level with year-conditioning.

- Prompt wording matters more than model choice.** Across prompts, r ranges 0.54–0.71; across models, 0.55–0.67.
- Cultural anchoring produces the largest shift away from the model's untimed guess.** Its predictions resemble the year-free baseline least ($r = 0.67$, vs. 0.78 for year alone).
- Only 29% of model-prompt-year combinations** exceed the null's 95% CI. Some knowledge is recovered, but recovery is far from uniform.
- Every model gains from conditioning, but in different amounts.** Mistral-7B improves most (+0.17) and Qwen3.5-0.8B least (+0.11), with the larger models in between. The gain does not scale with size.

5 Task 2: Prestige

No survey of occupational status exists before 1925, so there is no contemporary ground truth for most of this period. We score against seven historical classifications, coded to six ordinal tiers.



Probability tracks prestige tier. Mean probability per tier, 1850–1940. The top tier draws $3\times$ the probability of the next and $18\times$ the bottom. Separation is coarse: top vs. rest and non-manual vs. manual are clean, adjacent tiers are not.

+0.67 best recovery OLMo-2-32B	+0.33 worst model falcon-40b	18× top vs. bottom tier probability
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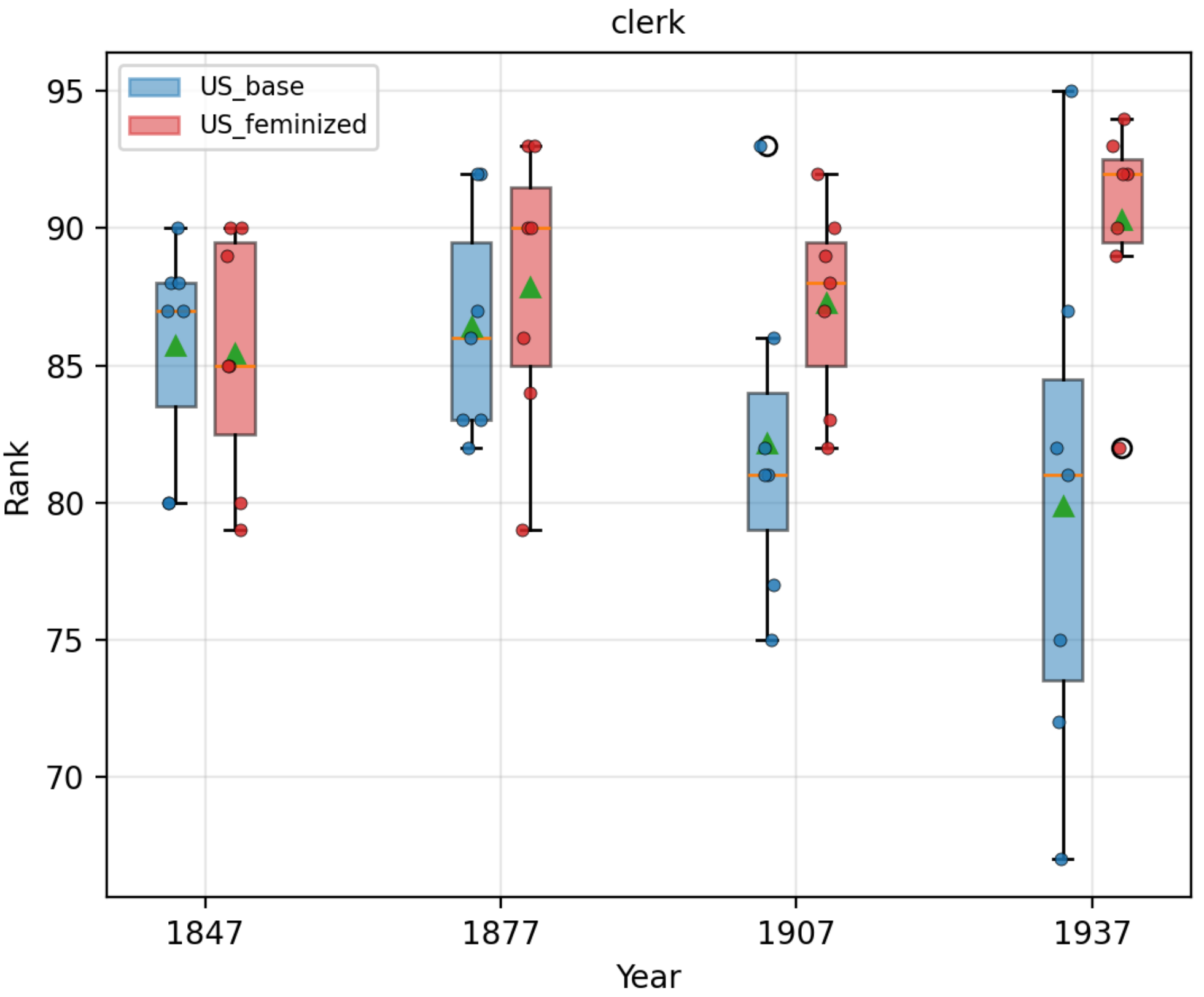
- Every model recovers at least some of the hierarchy.** Mean Spearman r ranges +0.33 to +0.67 across the seven models and is positive in all four years. OLMo-2 is best in every year.
- Size does not predict recovery.** falcon-40b is the weakest model in the set, below Qwen3.5 at 1/50 the parameter count.
- Anachronisms are suppressed.** All eight modern occupations fall below the bottom tier, with astronaut the most probable at 0.0007.
- Recovery is not uniform** Top- and bottom-tier occupations are recovered more accurately than mid-tier ones. This may be due to prompt design, but also reflects trends in historical sources.
- Recovery is stable.** Mean r by year moves little — +0.45 (1850), +0.51 (1880), +0.48 (1910), +0.52 (1940).

8 Future Directions

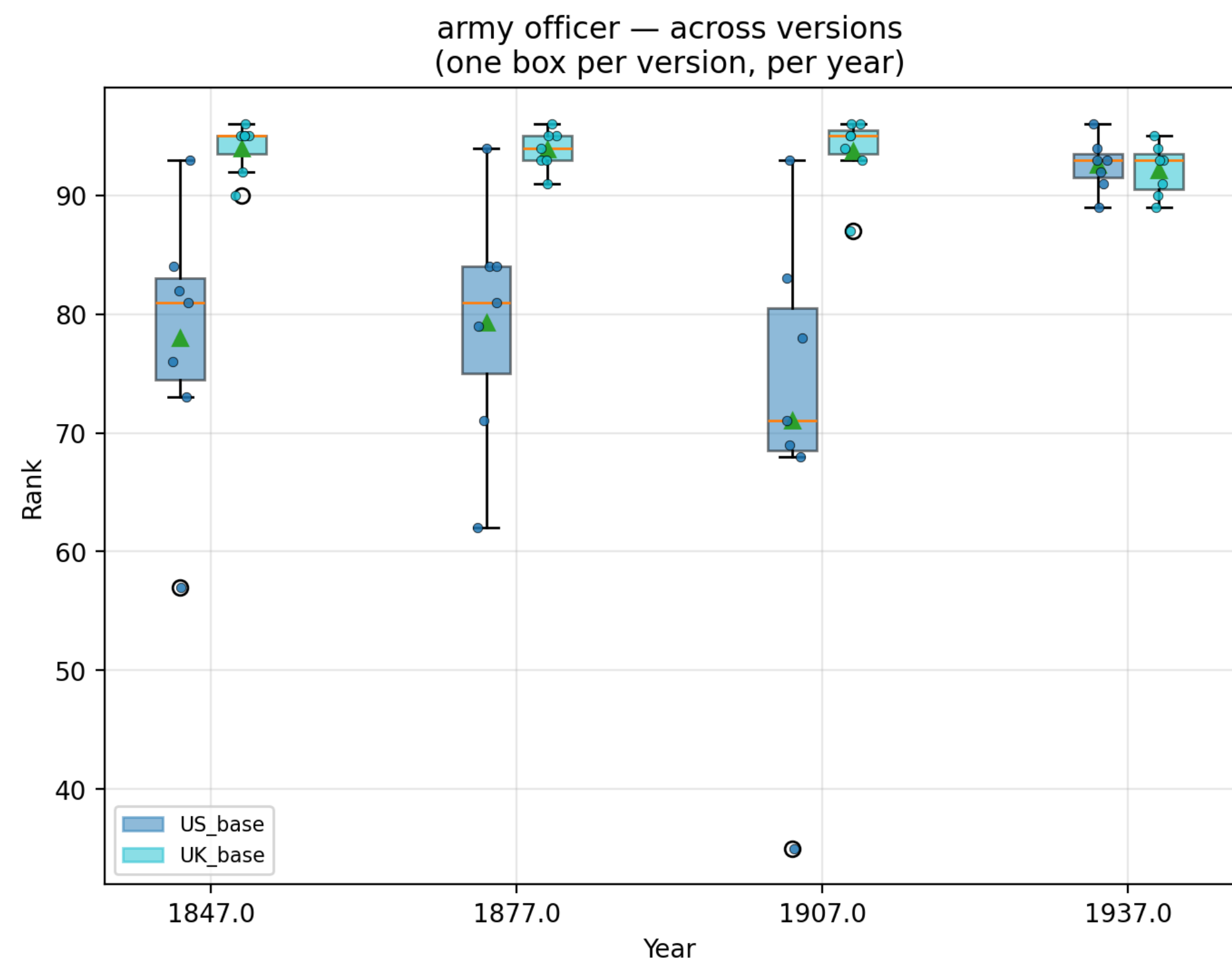
- Generating plausible counterfactuals.** Asking about events that never came to pass. By construction, counterfactuals have no ground truth, which raises the question of validation.
- Experimental and simulation-based histories.** Prompting makes the vantage point easy to manipulate; varying it systematically might approximate an experimental design.
- Vintage models.** As models trained on bounded historical corpora multiply, systematic comparison would show how well they recover period collective mentalities.
- Representational probes.** Hidden states may carry historical knowledge that does not reach the output.

6 Perspective moves

Changing the vantage point the prompt encodes changes the recovered hierarchy in ways a historian might predict. This makes shifts interpretable rather than noise.



Gendered vantage: clerk splits in two. Base prompt vs. feminized prompt, 1850–1940. Clerk trends downwards with the base prompt, but upwards with the feminized prompt.



Geographic vantage: army officer sits higher in Britain. US vs. UK prompt, 1850–1940. The UK runs above the US through 1910, then the two converge.

- One word change shifts recovery.** Only the vantage changes — a gender, then a country — and the trajectories diverge.
- Both splits match expectations.** Office work fell in general standing as it feminized, while rising within women's occupations as it displaced domestic service. British 19th century army officer corps were aristocratic in ways American counterparts never were. The convergence fits with WWII changing attitudes.
- Tentative.** These two carefully chosen examples show prompting *can* shift recovered perspectives. Whether it recovers these patterns in general is the next test.